# Learning Python

★ For each problem, give yourself **30 minutes total**: think independently for the first **10 minutes**; if no ideas emerge, check the LeetCode topics and hints and think for **10 more minutes**; if you still have no direction after **20 minutes**, read the solution.

- `Counter(iterable)` (from `collections`) maps each element to its frequency and is the go-to tool for frequency counting in interview problems.
- `most_common(n)` returns the $n$ most frequent `(element, count)` pairs in descending order (omit $n$ for all), making it ideal for top-$k$ frequency problems.
- `"".join([char for char in s if char.isalnum()])` strips a string to only letters and digits using `isalnum()`, which returns `True` for a–z, A–Z, and 0–9.
- `char.isnumeric()` accepts any Unicode numeric character (including fractions and superscripts), while `isdigit()` is stricter (ASCII digits 0–9 only).
- Never call `pop(i)` inside a loop that increments `i` forward, as each removal shifts remaining indices; a backward loop such as `for i in range(k, -1, -1): nums.pop(i)` is safe.
- `del nums[i]` is slightly faster than `nums.pop(i)` at the same O(n) complexity because `pop` has the extra overhead of returning the removed value, which can matter in tight loops over large lists.
- `list.index(i)` returns the index of the first occurrence of value `i` in the list, raising `ValueError` if it is not found.
- `list.count(value)` returns the number of times `value` appears in the list.
- `math.prod(list)` (from `math`) returns the product of all elements in the iterable, returning `1` for an empty iterable.
- `arr.insert(index, new_value)` inserts `new_value` at the given `index`, shifting all subsequent elements one position to the right.

- `my_dict.values()` returns a view object of all values in the dictionary, which can be saved into a list with `values_list = my_dict.values()`.
- `sum(my_dict.values())` returns the sum of all values in the dictionary, useful for quickly computing totals without an explicit loop.
- `max(my_dict.values())` returns the largest value in the dictionary, useful for finding the maximum frequency or count in a frequency map.
- `round(number, ndigits)` rounds `number` to `ndigits` decimal places (default 0), returning an `int` when called with no `ndigits` argument or a `float` otherwise.
- A **stack** in Python is simply a `list` where `append(val)` acts as push and `pop()` (no argument) acts as pop, both in O(1) time.
- Prefer `stack.append()` and `stack.pop()` over slice assignment such as `s = s[:-1]`, which creates a full O(n) copy each time and causes quadratic runtime in a loop.
- Two separate `if` statements in a loop are evaluated independently (both can trigger in the same iteration), whereas `if/elif` ensures only one branch fires per iteration.
- The **slow and fast pointer** technique (Floyd's cycle finding algorithm) uses two pointers where fast moves two steps per iteration and slow moves one; because their relative speed is one, they are guaranteed to meet if a cycle exists.
- To find the **nth node from the end** of a linked list, advance fast by $n$ steps first, then move both slow and fast together one step at a time; when fast reaches the end, slow is pointing to the node whose `next` should be updated.

# 1 LeetCode Practice

Table 1: LeetCode practice tracker

| Question | Level | Date | Revisit | Done |
|---|---|---|---|---|
| **1. Arrays & Hashing** | | | | |
| 217. Contains Duplicate | Easy | | | ✓ |
| 1. Two Sum | Easy | | | ✓ |
| 242. Valid Anagram | Easy | | | ✓ |
| 383. Ransom Note | Easy | | | ✓ |
| 349. Intersection of Two Arrays | Easy | | | ✓ |
| 1512. Number of Good Pairs | Easy | | | ✓ |
| 205. Isomorphic Strings | Easy | | | ✓ |

| Question | Level | Date | Revisit | Done |
|---|---|---|---|---|
| 49. Group Anagrams | Medium | | | ✓ |
| 347. Top K Frequent Elements | Medium | | | ✓ |
| 238. Product of Array Except Self | Medium | | | ✓ |
| **2. Two Pointers** | | | | |
| 344. Reverse String | Easy | | | ✓ |
| 125. Valid Palindrome | Easy | | | ✓ |
| 27. Remove Element | Easy | | | ✓ |
| 26. Remove Duplicates from Sorted Array | Easy | | | ✓ |
| 283. Move Zeroes | Easy | | | ✓ |
| 977. Squares of a Sorted Array | Easy | | | ✓ |
| 88. Merge Sorted Array | Easy | | | ✓ |
| 167. Two Sum II: Input Array Is Sorted | Medium | | | ✓ |
| 11. Container With Most Water | Medium | | | ✓ |
| 15. 3Sum | Medium | | | ✓ |
| **3. Sliding Window** | | | | |
| 643. Maximum Average Subarray I | Easy | | | ✓ |
| 1876. Substrings of Size Three with Distinct Characters | Easy | | | ✓ |
| 2269. Find the K-Beauty of a Number | Easy | | | ✓ |
| 1984. Minimum Difference Between Highest and Lowest of K Scores | Easy | | | ✓ |
| 1652. Defuse the Bomb | Easy | | | ✓ |
| 219. Contains Duplicate II | Easy | | | ✓ |
| 121. Best Time to Buy and Sell Stock | Easy | | | ✓ |
| 3. Longest Substring Without Repeating Characters | Medium | | | ✓ |
| 567. Permutation in String | Medium | | | ✓ |
| 424. Longest Repeating Character Replacement | Medium | | | ✓ |
| **4. Stack** | | | | |
| 20. Valid Parentheses | Easy | | | ✓ |
| 682. Baseball Game | Easy | | | ✓ |
| 1047. Remove All Adjacent Duplicates In String | Easy | | | ✓ |
| 1544. Make The String Great | Easy | | | ✓ |
| 844. Backspace String Compare | Easy | | | ✓ |
| 232. Implement Queue using Stacks | Easy | | ⋆ | |
| 496. Next Greater Element I | Easy | | ⋆ | |
| 155. Min Stack | Medium | | | ✓ |
| 150. Evaluate Reverse Polish Notation | Medium | | | ✓ |
| 739. Daily Temperatures | Medium | | | ✓ |
| **5. Binary Search** | | | | |
| 704. Binary Search | Easy | | | ✓ |
| 374. Guess Number Higher or Lower | Easy | | | ✓ |
| 35. Search Insert Position | Easy | | | ✓ |
| 278. First Bad Version | Easy | | | ✓ |
| 744. Find Smallest Letter Greater Than Target | Easy | | | ✓ |

| Question | Level | Date | Revisit | Done |
|---|---|---|---|---|
| 367. Valid Perfect Square | Easy | | | ✓ |
| 69. Sqrt(x) | Easy | | | ✓ |
| 74. Search a 2D Matrix | Medium | | | ✓ |
| 875. Koko Eating Bananas | Medium | | ⋆ | |
| 153. Find Minimum in Rotated Sorted Array | Medium | | | ✓ |
| **6. Linked List** | | | | |
| 876. Middle of the Linked List | Easy | | ⋆ | |
| 83. Remove Duplicates from Sorted List | Easy | | | ✓ |
| 203. Remove Linked List Elements | Easy | | | ✓ |
| 206. Reverse Linked List | Easy | | ⋆ | |
| 141. Linked List Cycle | Easy | Nov 9 | ⋆ | |
| 160. Intersection of Two Linked Lists | Easy | Nov 9 | ⋆ | |
| 234. Palindrome Linked List | Easy | Nov 10 | | ✓ |
| 19. Remove Nth Node From End of List | Medium | Nov 10 | ⋆ | |
| 2. Add Two Numbers | Medium | Nov 11 | | |
| 143. Reorder List | Medium | Nov 11 | | |
| **7. Trees (DFS and BFS)** | | | | |
| 94. Binary Tree Inorder Traversal | Easy | Nov 12 | | |
| 104. Maximum Depth of Binary Tree | Easy | Nov 12 | | |
| 226. Invert Binary Tree | Easy | Nov 13 | | |
| 100. Same Tree | Easy | Nov 13 | | |
| 101. Symmetric Tree | Easy | Nov 16 | | |
| 112. Path Sum | Easy | Nov 16 | | |
| 543. Diameter of Binary Tree | Easy | Nov 17 | | |
| 102. Binary Tree Level Order Traversal | Medium | Nov 17 | | |
| 98. Validate Binary Search Tree | Medium | Nov 18 | | |
| 230. Kth Smallest Element in a BST | Medium | Nov 18 | | |
| **8. Heap / Priority Queue** | | | | |
| 1464. Maximum Product of Two Elements in an Array | Easy | Nov 19 | | |
| 1046. Last Stone Weight | Easy | Nov 19 | | |
| 2558. Take Gifts From the Richest Pile | Easy | Nov 20 | | |
| 506. Relative Ranks | Easy | Nov 20 | | |
| 1337. The K Weakest Rows in a Matrix | Easy | Nov 23 | | |
| 2335. Minimum Amount of Time to Fill Cups | Easy | Nov 23 | | |
| 703. Kth Largest Element in a Stream | Easy | Nov 24 | | |
| 215. Kth Largest Element in an Array | Medium | Nov 24 | | |
| 973. K Closest Points to Origin | Medium | Nov 25 | | |
| 621. Task Scheduler | Medium | Nov 25 | | |
| **9. Backtracking** | | | | |
| 509. Fibonacci Number | Easy | Nov 26 | | |
| 231. Power of Two | Easy | Nov 26 | | |
| 326. Power of Three | Easy | Nov 27 | | |
| 590. N-ary Tree Postorder Traversal | Easy | Nov 27 | | |
| 257. Binary Tree Paths | Easy | Nov 30 | | |

| Question | Level | Date | Revisit | Done |
|---|---|---|---|---|
| 1863. Sum of All Subset XOR Totals | Easy | Nov 30 | | |
| 401. Binary Watch | Easy | Dec 1 | | |
| 78. Subsets | Medium | Dec 1 | | |
| 46. Permutations | Medium | Dec 2 | | |
| 39. Combination Sum | Medium | Dec 2 | | |
| **10. Graphs** | | | | |
| 1791. Find Center of Star Graph | Easy | Dec 3 | | |
| 997. Find the Town Judge | Easy | Dec 3 | | |
| 1436. Destination City | Easy | Dec 4 | | |
| 733. Flood Fill | Easy | Dec 4 | | |
| 463. Island Perimeter | Easy | Dec 7 | | |
| 1971. Find if Path Exists in Graph | Easy | Dec 7 | | |
| 993. Cousins in Binary Tree | Easy | Dec 8 | | |
| 200. Number of Islands | Medium | Dec 8 | | |
| 994. Rotting Oranges | Medium | Dec 9 | | |
| 207. Course Schedule | Medium | Dec 9 | | |
| **11. Dynamic Programming** | | | | |
| 70. Climbing Stairs | Easy | Dec 10 | | |
| 1137. N-th Tribonacci Number | Easy | Dec 10 | | |
| 746. Min Cost Climbing Stairs | Easy | Dec 11 | | |
| 118. Pascal's Triangle | Easy | Dec 11 | | |
| 119. Pascal's Triangle II | Easy | Dec 14 | | |
| 338. Counting Bits | Easy | Dec 14 | | |
| 392. Is Subsequence | Easy | Dec 15 | | |
| 198. House Robber | Medium | Dec 15 | | |
| 322. Coin Change | Medium | Dec 16 | | |
| 300. Longest Increasing Subsequence | Medium | Dec 16 | | |